

Data Flow Diagram

Data Flow Diagram Analysis

Date	14 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

1. Architectural Boundaries

The system operates as a secure three-tier application. Client interactions are proxied through a Node/Express backend so external API access and database transactions remain protected. The following data flows describe how information moves across the Personalized Networking Assistant.

2. Core Data Flows

DF1 (User → Client UI):

User inputs professional metadata, custom subscription budgets, or event descriptions.

DF2 (Client UI → Express Server):

HTTP POST/GET requests are sent to backend endpoints such as /api/auth and /api/suggestions/generate.

DF3 (Express Server → Wikipedia REST API):

Queries retrieve verified abstracts and desktop URLs to provide factual grounding.

DF4 (Express Server → Google Gemini Model):

A consolidated prompt containing the user profile, Wikipedia context, and history constraints is dispatched securely to gemini-3.5-flash.

DF5 (Gemini → Express Server):

A structured JSON payload containing curated suggestion categories is returned, validated, and processed.

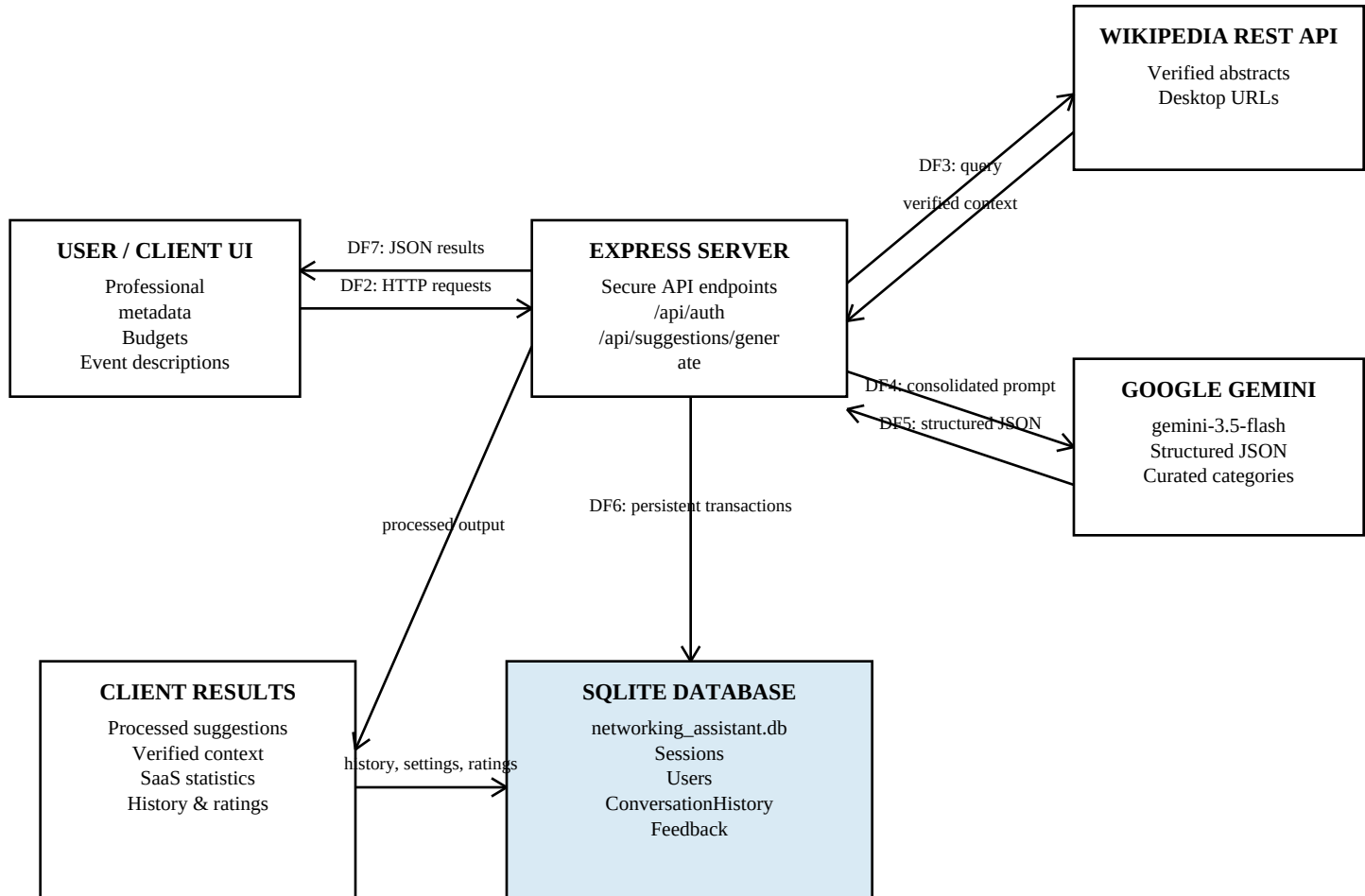
DF6 (Express Server → SQLite Database):

Historical logs, extracted topics, user settings, and ratings are stored in networking_assistant.db.

DF7 (Express Server → Client UI):

Processed suggestions and updated SaaS statistics are returned as JSON for frontend rendering.

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3. Data Store Definitions

- Sessions Table: Manages user login states.
- Users Table: Stores salt-hashed credentials, profession, goals, and customized plan parameters.
- ConversationHistory Table: Stores extracted topics, event descriptions, and structured JSON suggestions.
- Feedback Table: Stores ratings, likes, dislikes, and text comments for analytics dashboard rendering.